

Mobile app development is process of making software application that work on mobile devices such as smartphones & tablets.

These applications are designed to provide specific services that include entertainment, communication, Business solution or productivity tools for users.

Popular mobile OS:-

- 1) Android developed by Google.
- 2) iOS developed by Apple.

* Types of Mobile Apps:-

1) Native Application

- These applications are developed specifically for single platform.

e.g. Android based apps / iOS based apps.

Features of Native Apps:-

These apps offer high performance & better user application experience.

e.g. Android apps are developed using java, kotlin.

iOS apps developed using swift, objective C.

2) Hybrid Applications:-

These applications are built using web technologies such as HTML, CSS & JavaScript.

These application can run on multiple application platforms.

3) Cross Platformed Application:-

These application are developed once & developed on multiple platforms.

Popular frameworks:-

Flutter, React Native.

4) Progressive Web Application:-

These are web application that behave like mobile apps.

These applications can be accessed from browser without installing from app store.

* Mobile Application Development Lifecycle
- It is a structured process used to design, develop, test, deploy & maintain mobile Application.

- It includes different phases:-

i) Requirement, Gathering & Analysis

- This phase includes identifying business goals & target users.

- Define App Features & Functionality.
- Analyse competitions & market trends & choose the platform.

ii) Planning

- In this phase we need to estimate cost, time & resources.

- Here we identify the technologies, frameworks & tools.

- We also create project schedules & milestones.

iii) UI-UX Design

- In this phase we decide users interface & experiences.

- Here we create wireframes & prototypes.

- We also define navigation flow & visual designs.

iv) Development

- Here we implement application functionality.
- We develop functional frontend i.e. mobile interfaces.
- We also develop backend & servers.
- At include database also.
- Here we include 3rd party services if required.

v) Testing

- Here we verify application functionality, performance, security & usability.
- We identify & fix defects.

vi) Deployment

- Here we publish app to app stores.
- We also configure production environments.

vii) Maintenance & Support.

- Here we monitor app performance.
- We fix bugs & security issues.
- We release updates & new features.
- We also gather user feedback for improvements.

Requirement, gathering

↓
Planning

↓
UI/UX design

↓
Development

↓
Testing

↓
Deployment

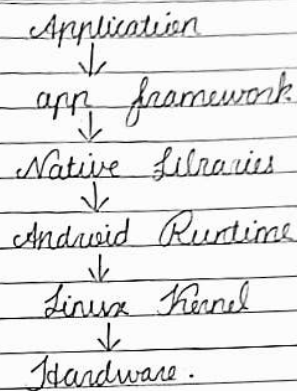
↓
Maintenance.

* Introduction to android.
Android is an operating system developed by Google for smart-phones & Tablets.

Android architecture:

- Android consist of several layers.
- 1) Linux Kernel:-
- The kernel consist of hardware abstraction layer, device & memory & process management utilities.
- 2) Android Runtime (ART):-
- It executes applications.
- It includes java & Kotlin libraries
- 3) Native Libraries:-
- It includes SQLite, open GL, media framework.
- 4) Application Framework:-
- Activity Manager
- Window Manager
- Content providers
- Notification managers.
- 5) Application Layer:-
- User installs apps.

5)



Structure of Android Program

An android application is mainly build using java/kotlin & XML layout files.

Structure of program is -

- 1) Package declaration -
It defines the package name of application.
- 2) Import statements -
It imports required android libraries
- 3) Main activity class -
It is main This is main screen of application.
- 4) On create method -
This method is 1st method called when an activity starts.
- 5) Layout File:
It defines all the user interface it is an xml file.

Every android contains 2 files xml called activity_main.xml. It contains all user interface. Layouts are stored in folder called res folder.

- 6) Main activity.java -
Here we write the program logic.
In addition to these files android also contains android.manifest.xml.
- This xml file contain all app information & user permission.

~~Android manifest.xml~~
~~add configuration & permission~~
~~java & kotlin files~~

Project structure.

Android Project

- AndroidManifest.xml.
 - ↳ app configuration, permission, activities
- Java / Kotlin Files
 - ↳ Main activity.java (or Main Activity.kt)
- res (Resources)
 - ↳ layout
 - ↳ activity-main.xml.
 - ↳ drawable
 - ↳ Images & Icons.
 - ↳ values
 - ↳ strings.xml
 - ↳ colors.xml
 - ↳ themes.xml
- ↳ gradle scripts
 - ↳ Build configuration

Listeners in Android.

listens to events.

Listener in Android waits for a user action & responds when the event occurs.

listeners are commonly used to handle button touches, clicks, key presses & other user interaction.

Common Type of Listeners

- 1) On click listeners:-
 - handles button click.
- 2) On long click listeners:-
 - handles long presses.
- 3) On touch listeners:-
 - handles touch events.
- 4) On key listeners:-
 - handle key presses.
- 5) On focus change listeners:-
 - It detects focus change.

Q Program to print your name on click of your button.

```

main Activity class {
    onCreate() {
        button.set OnClickListener() {
            textView.setText("Sherlock");
        }
    }
}

```

Widgets in Android.

In android widget is UI component that user can see & interact in an app.
e.g.

TextView - Display text
edit text - allows user to type text.

Button - Perform action when clicked.

ImageView - Display image

Checkbox - Lets user to select/deselect option.

RadioButton - Lets user choose one option from group.

Switch - Turn setting off/creating button using theme.

code:

<Button

android:layout width="wrap_content"

android:layout height="48"

android:text="Submit"/>

< Edit Text

```

android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:text="S" />

```

Edit text Textview
 Allows user input - ~~Deser~~ Like a label.

Code

```

package com.example.myapplication;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    TextView Name;
    Button btnShow;
    @Override
    protected void onCreate(Bundle

```

```

savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    txtName = findViewById(R.id.txtName);
    btnShow = findViewById(R.id.btnShow);
    btnShow.setOnClickListener(view -> {
        view -> is called a lambda expression in Java. It is a shorter way of writing an onclicklistener.
        txtName.setText("hello world");
    })
}

```

Explanation

- The program prints hello world in click of button.
 - Inside main activity I you define a text view object & button object.
 - Inside onCreate function we need to bind variable like object you created in layout file using function findViewById().

Syntax of binding variable is -
variable (name = findViewById(R.id.
object name));

- Next you have to set the
onClickListener of the button

Syntax:-

btnShow.setOnClickListener(view->{

Q. Write an android program in
java with 2 buttons button 1 &
button 2.

Q. Write a program to change the
text of button on click
of button 1.

→ 1) Button but1, but2;
but1 = findViewById(R.id.
Button1);

but2 = findViewById(R.id.
Button2);

but1.setOnClickListener(view->{
but2.setText("Jack");
});
}

Q1) Write an android program to
bring name & address address
in textview when you click
button.

2) ——— to find sum of
2 numbers with click of
button

Toast() Function:

- An android toast is a small pop up message that appears automatically.
- It is used to display brief information.
- Syntax
`Toast.makeText(context, "message",
 Toast.LENGTH_SHORT).show()`

Context:-

- Context is where you have to display message & usually it will be MainActivity.this

Message:-

- Message is text you want to display.

Toast.Length.Short

- It displays message for short duration
- You can also give toast.Length.LONG.
- That displays message for

longer duration.

The show function shows the message.

- Q Create/Write android function to display toast message good morning for short duration. Call this function in button click.

→

```
void displaymessage() {
    toast.makeText(this, "good morning",
        Toast.LENGTH_SHORT).show();
}

b1.setOnClickListener() {
    displaymessage();
}
```

- Q Define a function a toast message your Name, call function button click. The message should be shown for long duration.

→

```
void displaymessage() {
    Toast.makeText(this, "Jack Reacher",
        Toast.LENGTH_LONG).show();
}

b1.setOnClickListener() {displaymessage();}
```

* Checkboxes & Radio Buttons

Checkboxes is a ui component that allows the user to select or deselect an option.

It has 2 states:- checked. It is selected state

i) unchecked:- Non selected state is called unchecked

- Checking the state of checkbox. We can check the state of the checkbox by using isChecked function

isChecked() function returns true if the checkbox is selected else return false.

Radio Button:- is an ui interface. Allow to select only one option from a group

Difference between checkbox & Radio button is -
checkbox allows multiple selection

while radio button allow only single selection.

* Media Player

- is an android class used to play audio & video in android application.

- We can play media from local resources that are stored in raw folder.

- File store in the device or online streams.

- The methods of media player.

i) create():- MediaPlayer mdt;
To create object of media player.

ii) Start():-
To start playing the media

iii) Pause():-
To pause playing the media.

iv) Stop():-
To stop

v) seek To():-
Moves to particular location

vi) is playing():-

To check whether media is playing.

vii) release():-

To release all media player resources.

```
MediaPlayer md1;
md.create(this, "filename");
md.start();
md.play();
md.pause();
md.stop();
md.release();
```

* Menus in Android.

Menu in android is a collection of options or actions that allows users to perform different operations.

Types of Menus in Android

- In Android there are 3 types of menus.

1) Options menu:-

Provides actions that affect the entire application.

Usually it appears in the app bar or action bar.

- e.g. search, settings, save, delete etc.

2) Context Menu:-

- Appears when the user performs a long press on a particular view or object.

e.g. copy, delete, edit, share, etc.

3) Pop up menu:-

- Displays small menu when the user

right clicks / selects an object
such as button or image view

Difference between menu types.

Menu	purpose	appears
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i) Options -	general activity actions	- app bar, action bar.
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ii) Context -	actions on a particular view	- appears on long press
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iii) Pop up -	small list of actions	- appears mainly on click or right click.
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